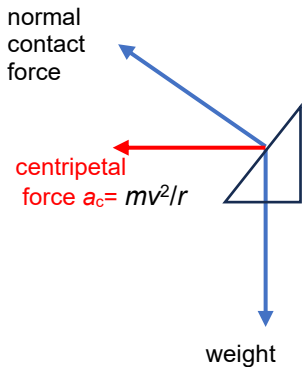


8

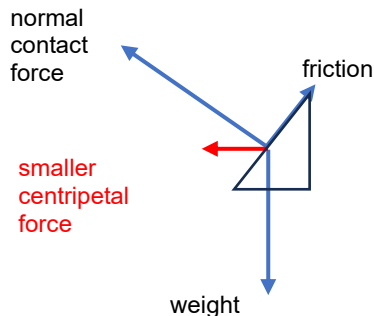
B

The required centripetal force is the vector sum of the horizontal component of friction and the horizontal component of normal contact force.

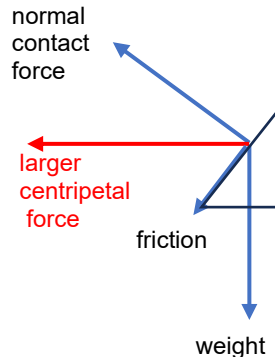
At v



below v



above v



When the car is moving below v , the required centripetal force (which is acting horizontally) is smaller; thus, frictional force should act upwards along incline to oppose the horizontal component of the normal contact force as shown below.

When the car is moving above the mentioned speed, the required centripetal force (which is acting horizontally) is larger; thus, frictional force should act downwards along incline to assist the horizontal component of the normal contact force as shown.

2023

HWA CHONG INSTITUTION (COLLEGE SECTION)

C2

Q	A	The gravitational force is the only force acting on the satellite. This force is providing the
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